

Licensing issues in a FOSS+cloud world

In March 2007, Linux Magazine published an article that revealed the loopholes associated with delivering free software over a network. The article, *The GPL Has No (Networked) Future* (<http://www.linux-mag.com/id/3017/>) pointed out that free software licence requirements to release source code are all triggered by the act of distribution, and that Web applications, which are not actually 'distributed' are therefore not bound by these licences. Other experts continued to point out this loophole.

In November 2007, the Free Software Foundation (FSF) released the Affero General Public License, a version of GPLv3, which intended to close the legal loophole associated with free software designed to be run over a network. The relevant clause read: "...if you modify the Program, your modified version must prominently offer all users interacting with it remotely through a computer network (if your version supports such interaction) an opportunity to receive the Corresponding Source of your version by providing access to the Corresponding Source from a network server at no charge, through some standard or customary means of facilitating copying of software."

Despite the rising number of cloud offerings with licences like AGPLv3, Apache, BSD and Creative Commons, apart from the lesser-known but equally good ones like the MIT Open Source License, there continues to be a lot of confusion concerning the implementation of FOSS over a cloud. We spoke to free software evangelist and consultant Niyam Bhushan about this...

From the point of view of free/open source software, is cloud computing good or bad?

Cloud computing is amoral. The licences and privacy policies you choose to implement decide whether you are creating a cloud-based golden cage, or a door that opens to the heavens of freedom.

Are there any real issues with the implementation or deployment of free/open-source software over a cloud? If so, is the problem solely faced by public clouds or also by private ones?

The problems of FOSS implemented over a cloud are manifold, if you love and honour digital freedom.

1. With desktops, the problem was relatively less acute, as your data was still on your local hard disk. With clouds, both the software and your data is locked away in the cloud. At any time, the powers that be can block or delete your data, or deny you access, by accident or design. A Flickr user was horrified to discover, one fine morning, that his entire archive of photos was deleted by Flickr due to an error on Flickr's part.
2. With Web-based software engines, you do not know whether the software obeys you, or another master. There could be hidden features, leaks, insidious code that could damage your business, work, or even your reputation. Every Web cloud and engine starts on the premise that the end-user should trust it. This trust could well be a trap.
3. The issues of privacy, in the name of collaboration, and the use and abuse of your data to build a profile of you are indeed alarming. There is no control over how that data could be used.
4. If data and software are Web-hosted and served globally, what legal jurisdiction is applicable to it is an old debate that may find new wrinkles.
5. Data licensing can also be another issue, depending on the type and purpose of the data. For that, a user may wish to have choices from among several Creative Commons licenses.

So, if some people wish to use a free/open-source licence for their software, and also wish to deliver it over a cloud, would they face any problems?

None whatsoever, if you use Affero GPL, also known as AGPLv3.

However, if someone wishes to use FOSS but not use AGPLv3, that too is possible. That is the freedom guaranteed by other copyleft licences, including GPLv2 and GPLv3. However, the spirit of freedom is not truly honoured for the end-user. It is only applicable to the individual or organisation deploying free software enabled as a service. This can be quite tragic for those who love digital freedom.

What are the best free/open-source licences for the FOSS+cloud combo?

AGPLv3.

Elastra and 3Tera are other similar offerings.

Nimbus is another notable tool-kit, maintained by the University of Chicago, which enables you to swiftly convert your cluster into an IaaS cloud. Currently, Nimbus is deployed into a Globus 4.0.x Java container, a system built around the Apache Axis engine. It supports three sets of remote interfaces: the Amazon EC2 Web services definition/description language (WSDL), the Amazon EC2 Query API, and the grid community Web services resource framework (WSRF). It also manages the security for these interfaces. The storage implementation is compatible with the Amazon S3 REST API, and virtualisation is based on Xen and KVM. Nimbus is known for being highly configurable and extensible.

George Paul brings to our notice another interesting cloud-computing project, Reservoir, funded by the European Union and coordinated by IBM. Last September, the group released its cloud stack featuring the Claudia Service Manager, a tool for automatic management of service scalability, and the OpenNebula Cloud Toolkit. OpenNebula is an open source cloud-computing tool-kit capable of managing several thousand virtual machines, along with large storage and networks. It supports all common cloud interfaces, and can fit into any existing data centre to help build a private, public or hybrid cloud.

Some of the other notable open source tools that are very useful in a cloud environment include Apache's Traffic Server (a fast, scalable and extensible HTTP/1.1 compliant caching-